Assignment 3 Instructions

PSYC417

Assignment 3

This assignment is going to test your skills in **data wrangling, visualization, function writing, and loops**. You may get to the answer **using any of the techniques we specify in class, unless the question requests otherwise. If so, you must use that technique.** Do not use techniques or code we didn't learn in class (e.g., from a package we haven't used).

Description

The data for this assignment (located in the **Datasets folder**) come from a study examining how different types of **background sound** affect listeners' ability to comprehend spoken stories. Participants listened to a series of short stories while one of four background conditions played simultaneously:

- `Music`: Jazz music played in the background.
- `ReverseMusic`: The same jazz music played in reverse (by a time-reversed version of the original music file).
- `Babble`: multi-talker speech babble (9 people speaking at the same time so that it sounds like noise).
- `Silent`: No background sound.

After listening to the short stories, participants answered questions about the stories. We will not be replicating the analyses in the original paper. In this snapshot, the primary numeric variable of interest is the `Total_BMRQ` (the total score on the **Barcelona Music Reward Questionnaire**, a measure of music reward sensitivity). We will be working with a number of other variables that were collected during the study (e.g., age, musicianship).

The datasets (again, in the **Datasets folder**) we are going to use are called: `study_demographics.csv` and `study_result.csv`. Load these into objects called demographics and results before beginning the assignment.

Questions

Write each question/number into the Quarto document as text and place any code or text to answer the question on the lines below it. In short, it should look something like below:

Question 1 - Text from question 1. (MAKE THIS ONE OF THE HEADER STYLES SO IT'S EASIER TO READ.)

Code chunk, if applicable.

Text answering question, if applicable.

Question 2 - Text from question 2. (MAKE THIS ONE OF THE HEADER STYLES SO IT'S EASIER TO READ.)

…. and so on.

Question 0 (5 pts)

Create a Quarto document titled "Assignment 3" with you listed as the author (1pt). **Make sure the document type is .docx (1pt).** When uploading the assignment to ELMS, ensure you upload both the .qmd and .docx files (3pts). (No fixing this after the deadline if you forget.)

Question 1 (25 pts)

Oh no! Our demographics data set does not contain age, which could be an important confound given our data. Instead, we have the date that the participant took the survey and the day they were born.

Using base R, create a function that solves this problem. The function should take M/D/Y formatted data as inputs like we have. Here is some code to help you convert such data into something R can work with (i.e., Date data). Note that m and d are LOWERCASE in the format argument and Y is UPPERCASE.

```
as.Date("11/03/2017", format = "%m/%d/%Y")
```

```
[1] "2017-11-03"
```

If you calculate the difference of two Date data points in R (e.g., as.Date(…) - as.Date(…)) it will give you a character expression. Coercing this to numeric will give you the number of DAYS between the two points.

```r
example = as.Date("2017-11-04") - as.Date("2017-11-03")
print(example)
```

```
Time difference of 1 days
```

```r
as.numeric(example)
```

```
[1] 1
```

Name your function **age calculator** and give it **two** arguments**: take_date** (when they took the survey) and **birth_date** (well, when they were born). Have the function return the number of YEARs between the two points. (You will have to do something to convert the output above from days to years.)

Once you've written this function and verify it works, add a column to the demographics data frame called **age** and use your function to create it. In the demographics data, **date** is the day they took the survey and **dob** is the year the participant was born.

Question 2 (20 pts)

Since the authors conducted a **within-subjects** study, the data are in long format. We will need to work the data into a form we are familiar with. We will need to make the following adjustments: 1.) only show cases where Music was the Background Type. This will reduce the number of rows per participant to two (since they had two different SNR levels (Minus 2 and Plus3). Next, group the data frame by participant ID and calculate the average accuracy and BMRQ score. Store these adjustments into a data frame called **`short_res`**, where there should now be one row per participant.

Note: You can do this all in one pipeline if you use the tidyverse.

Note: You do not need to retain the Background_Type or SNR columns. We just need `participant`, `accuracy`, and `Total_BMRQ`.

Question 3 (20 pts)

Join `short_res` and `demographics` data frames. (Think about which columns share information between the two data frames.) We do not want to lose any rows in this join, so choose a type of join where nothing is excluded. Name the final data frame `full_data`.

Finally, remove the `date` and `dob` columns, as well as any column that starts with **freq** or **pref**.

Question 4 (15pts total)

Create the following three plots:

Q4.a (5 pts)

in theme classic: 1.) Create a stacked bar graph for the count of attention disorder filled by gender, 2.) Make the outlines of the bars black, 3.) Make sure the bars are side-by-side instead of on top of one another.

Q4.b (5 pts)

In theme black and white: 1.) Create a histogram with 20 bins for accuracy, 2.) Make the bar lines black areas under the curve colored by whether they were a musician, 3.) Make the colors forestgreen and #445566, respectively. Label the x-axis **X** and label the y-axis **Y**.

Q4.c (5 pts)

In APA format: 1.) Create a scatterplot for `accuracy` by `Total_BMRQ`, 2.) make it a faceted grid with `musician` as the rows of the grid and `gender` as the columns, 3.) color the points by the age of the participants, and 4.) give the overall plot a title called "*Complicated Scatterplots*".

Question 5 (25pts)

Create a function called `Pplots` that computes a scatter plot in base R (i.e., the plot() function) for every pair of **NUMERIC** variables in a data frame. APA You must use a loop to answer this question and it should generalize to a data frame with **any number of columns**. Do not have it generate an error if there are non-numeric variables; just skip over them. Once you've written the function, you can verify with the pairs() function that your new function works.

Hint: Outside of the loops and conditionals, there is only one line of code needed to solve this problem.

Bonus challenge a (0.05 pts)

in ggplot2 Make the points blue and add a regression line with no standard errors bar in each plot.

Bonus challenge b.) (0.05 pts)

Have the function skip all redundant plots. (e.g., If column x is plotted with column y, do not then plot y by x.)

Bonus challenge c.) (0.05 pts)

Modify the function such that it will choose the appropriate plot type based on the combination of variables (scatterplot for two numeric, boxplot for numeric and categorical, etc.).

The End

Render the document and upload the .qmd and .docx files to ELMS.

Thank you! :)